PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis
AUTHORS	Ma, Jiang; Liu, Xiaoxiao; Lu, Huaimin; Zhang, Di; Zhao, Tianyu; Wang, Ju; Jin, Song

VERSION 1 – REVIEW

REVIEWER	Fernando Domínguez-Navarro University of Nottingham
REVIEW RETURNED	15-Sep-2021

GENERAL COMMENTS	Thanks for the opportunity to review the manuscript titled “Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis”. I consider the topic of the study to be interesting and potentially relevant, due to arthroscopic partial meniscectomy (APM) is a high-volume surgery, and the economical and health consequences for the public health are elevated. However, some aspects of the present manuscript need to be improved. First, there are some methodological issues that need to be clarified (inclusion criteria and outcomes assessed). Moreover, the introduction and discussion sections do not reflect properly all the relevant aspects regarding rehabilitation after APM: (I) sensorial and functional limitations after APM, (II) the relationship between degenerative meniscus tear and knee osteoarthritis, (III) the role of exercise therapy, and proprioceptive training in particular, in the rehabilitation after APM, (IV) discuss deeper whether it is necessary to add proprioceptive training to the “usual” rehabilitation after APM. Do patients need it? Or are just strengthening exercises enough? Lastly, I consider the English writing level to be suboptimal. More detailed feed-back is shown below: ABSTRACT Line 27: Typographic mistake. There should be a space between characters INTRODUCTION Page 3 Line 13: I would omit this sentence. It does not seem to contribute relevant things to the concern of the study Line 27: It might also be mentioned that this type of injury should be understood in the context of knee OA.

	Line 29: Authors should emphasize the high volume of this surgery Line 33: I suggest a deeper look into the functional and muscular deficits that patients who undergo APM often exhibit in short and long-term after surgery Line 47: The authors could make an effort to summarize more specifically which are the effects of proprioceptive training. Does evidence say that it is an intervention that could be applied as a single treatment for that condition? Is it better in combination with other interventions? What outcomes are expected to improve with this intervention? Line 47: In addition, paragraphs one and two should be better connected. What deficits do those patients normally exhibit and why proprioceptive training may be the key to ameliorate those deficits? Please, explain better. Lines 48-55: In my opinion, the need for this study could be better justified. Why may be proprioceptive training an appropriate intervention after meniscus arthroscopy? What are the expectations? How and when should it be performed? Page 4: Line 7: Topic #4 of the PRISMA Checklist is not clearly shown in the manuscript. Please, define better the objective according to PIRSMA checklist description METHODS Line 29: Typographic mistake Line 45: Proprioceptive training may be also named "sensorimotor training". This keyword should be included in the search strategy. Page 5: Line 5: Not really clear what the population inclusion criteria are for the present systematic review. As it is explained, it is not clear if it includes subjects who had undergone surgery for meniscus injury, or if it includes subjects who were operated for knee problems, and after that surgery, had developed meniscal injury. In addition, in the introduction section, you correctly mentioned that there are two types of meniscus injury: one in young athletics subjects (traumatic tear) and the other in middle-aged subjects (degenerative tear). However, when you explain the population inclusion criteria, you do not specify which type of meniscus injury you are including. Considering the important differences that may exist in age, physical function and comorbidities between these two cohorts of subjects, it is not appropriate to make just a single analysis for these two groups. If you only include degenerative tears, please, clarify. If you combine both, make separate analyses. Line 7: Please, explain better which intervention performs the control group. “Convectional care” may be misleading. Please, also clarify that control group intervention do not include proprioceptive exercises

	Line 3-25: Item # 6 of the PRSIMA Checklist is not fully done. Please, clarify this topic Line 33: At the beginning of the sentence, it is better to write the number in words, instead of figures Line 47: Is it better if the manuscript avoids first or second person It is not well explained what outcomes have been evaluated, RESULTS Page 6: Line 49: It is better if you use the term “potentially relevant articles”, due to the fact that you should further assess if they are finally relevant Page 7: Line 19-25: In the methods section you mentioned that two researchers independently assessed the quality of the study using PEDRO scale. However, in Table 3 only appears a column of punctuations. Therefore, it is unknown which is the punctuation given by each of the researchers and if there was any disagreement between them. The same for risk of bias assessment. Lines 56-58: This sentence is not clear. Could authors rephrase it please? Page 8: Line 37: Clarify which tools have been used to test knee function. Are self-reported questionnaires or physical performance-based? Do all studies use the same approach? The selection process of the articles included in the review (Figure 1) is not in accordance with the inclusion and exclusion criteria exposed in the ‘methods’ section. You have excluded 1 record because “about anterior cruciate ligament injury” but there was no mention about it in ‘methods'. Furthermore, it is not clear why you have excluded 5 articles “about meniscus injury”. Please correct it. DISCUSSION As a summary, the discussion section does not reflect the important points of the study (more explained in the commentaries below). Therefore, I suggest that authors should rephrase this section. I can read in the manuscript a lot of information (even maybe too much) about what is proprioception and how it works, but I miss more information regarding to how proprioception is affected in a meniscus injury and what is the impact of arthroscopic partial meniscectomy on proprioception. I consider that this review is supposed to be seeking if including proprioceptive training to conventional rehabilitation may improve

	knee function (both physical function and self-reported) over just conventional rehabilitation. However, this question is not resolved in the discussion. It is not discussed if proprioceptive training may have clinical and functional implications on patients undergoing APM. I suggest to briefly compare the results obtained to other studies reviewing the effects of including proprioceptive training to conventional rehabilitation in other knee injuries Although it is little mentioned at the end of the limitations paragraph, you should discuss more about the dose, time, intensity, and type of exercises included in the proprioceptive interventions across all studies. Is it heterogenous? Do they lack some progressions? Do they do “enough” sessions?

REVIEWER	Keng Lin Wong National University of Singapore
REVIEW RETURNED	21-Oct-2021

GENERAL COMMENTS	Overall, an important paper to discuss about post operative physiotherapy. The authors have did their best to consolidate the sparse literature, however, due to the limited amount of studies, the meta-analysis unfortunately is difficult to be considered as significant with the high I2. Meniscus repair versus partial menisectomy is likely the cause of the reason why the studies were so heterogenous. I would suggest: 1) network meta-analysis of meniscus repair only or partial menisectomy only 2) if the results are too sparse then concentrate on the systematic review aspect only and present both isolated meniscus repair or partial meniscectomy

REVIEWER	Abdelrahman Abushouk Ain Shams University
REVIEW RETURNED	17-Nov-2021

GENERAL COMMENTS	There are three major limitations that affect the reliability of the current meta-analysis findings. While the authors did attempt to mitigate them, their effects still apply. 1. The small number of included studies, which also have a small number of patients. 2. The high heterogeneity observed across most analyzed outcomes. Rightfully, the authors used random effects model and applied sensitivity analysis, when needed. 3. The additional major limitation is the different controls in the included studies, possibly explaining the heterogeneity observed in some outcomes. While some authors combine studies with different controls, it is not advised to do that and even when done, should be supplemented by subgroup analysis according to control type. I still have few comments: 1. In the methods section, the outcome measurements and definitions must be added, so that the readers understand what the results mean.

	2. Also, it is imperative to define the intervention and mention how it was conducted in different studies, better done in a table. 3. fixed effect NOT fixed effects. 4. Page 6, line 37: What does it mean: The level of meta-analysis was set at .... Do you mean p value of significance? 5. The heterogeneity should better be interpreted according to Cochrane handbook cut-offs. https://handbook-5-1.cochrane.org/chapter_9/9_5_2_identifying_and_measuring_heterogeneity.htm 6. Considering the limitations mentioned earlier, the authors are advised to perform GRADE assessment to evaluate the certainty of evidence. 7. Flexor Muscle Strength outcome: The significance here is also marginal (p= 0.05). Even if considered significant, its clinical value is questioned. 8. Page 8, line 27: mention the exact p value. 9. The figures for each outcome (before and after reducing heterogeneity) should be combined, making it easier for the reader. 10. In the "Evaluation of Knee Functions" outcome, the authors performed sensitivity analysis by removing two studies at a time. Not sure if this methodologically valid.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Fernando Domínguez-Navarro, University of Nottingham

Comments to the Author:

Thanks for the opportunity to review the manuscript titled “Effects of proprioceptive training in the recovery of patients submitted to meniscus surgery: systematic review and Meta-analysis”. I consider the topic of the study to be interesting and potentially relevant, due to arthroscopic partial meniscectomy (APM) is a high-volume surgery, and the economical and health consequences for the public health are elevated. However, some aspects of the present manuscript need to be improved.

Response: thank you for your comments concerning our paper. These comments are all valuable and very helpful for revising and improving our paper. We have read comments carefully and have made correction which we hope meet with approval.

First, there are some methodological issues that need to be clarified (inclusion criteria and outcomes assessed). Moreover, the introduction and discussion sections do not reflect properly all the relevant aspects regarding rehabilitation after APM: (I) sensorial and functional limitations after APM, (II) the relationship between degenerative meniscus tear and knee osteoarthritis, (III) the role of exercise therapy, and proprioceptive training in particular, in the rehabilitation after APM, (IV) discuss deeper whether it is necessary to add proprioceptive training to the “usual” rehabilitation after APM. Do patients need it? Or are just strengthening exercises enough?

Response: according to your comments, we have revised a lot of contents in the introduction and discussion section, and the span of revision is also large. We have not provided a complete reply for this part. Therefore, we suggest you to read the revised manuscript.

The inclusion criteria and outcomes assessed section.

Eligibility criteria for this review based on PICOS frameworks (Population, Intervention, Comparison, Outcome and Study) were as follows: (1) patients who have undergone APM for meniscus injury caused by traumatic tear, whose race, nationality, and duration of disease are not limited, patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded; (2) Interventions: proprioceptive training alone or combined training with conventional rehabilitation were seemed as experimental groups; (3) conventional rehabilitation training (including ankle pump movement, continuous passive motion, weight bearing exercise, strength training, etc.) were applied to control groups, control groups

intervention did not include proprioceptive training; (4) at least one of the outcomes of extensor, flexor muscle strength, knee function score or proprioception test was reported; (5) randomized controlled trial (RCT) published in English or Chinese.
The introduction and discussion sections.
(I) postoperative patients experience pain and swelling leading to loss of range of movement, proprioceptive deficits, neuromuscular and biomechanical changes, decreased quadriceps femoris muscle strength, and further leading to dysfunction. The dysfunction especially in proprioception which perceives changes in the state of motion and initiates protection and muscular reflexes to keeps dynamic stabilizing are prone to increase the risk of degenerative disease or reinjury in long-term after surgery
(II) The other is defined as a degenerative change and often accompanied by degenerative changes in cartilage, mostly found in middle-aged and elderly people, which often involved in knee osteoarthritis.
(III) (IV) Proprioception plays an important role in postoperative rehabilitation. The inputs from the sensory organs are processed in the brain and integrated with visual and vestibular information to generate a sense of position and movement through space. Proprioceptive training can stimulate the sensory organ to send specific signals to control the relevant muscles to maintain stability through specific dynamic movement exercises such as practices of balancing, positioning, gait flexibility, agility and neuromuscular.15-17 In recent years, adding proprioceptive training to conventional rehabilitation has been widely used in postoperative rehabilitation of joint at different times, some on the first day after surgery, and some three months after surgery, which proved to be effective in improving proprioception and activity function.

Lastly, I consider the English writing level to be suboptimal. More detailed feed-back is shown below:
Response: according to your comments, we carefully corrected words and phrases in our manuscript and our manuscript have been checked by someone familiar with writing article in English.

ABSTRACT

Line 27: Typographic mistake. There should be a space between characters
Response: according to your comments, we have made the modification and checked the full text.

INTRODUCTION

Page 3

Line 13: I would omit this sentence. It does not seem to contribute relevant things to the concern of the study
Response: according to your comments, we have deleted this sentence.

Line 27: It might also be mentioned that this type of injury should be understood in the context of knee OA.
Response: according to your comments, we have supplemented that the other was defined as a degenerative change of the cartilage, mostly found in middle-aged and elderly people, which often involved in knee osteoarthritis.

Line 29: Authors should emphasize the high volume of this surgery

Response: according to your comments, we have supplemented that study showed that about 636 000 knee arthroscopy procedures are performed every year in the United States.

Line 33: I suggest a deeper look into the functional and muscular deficits that patients who undergo APM often exhibit in short and long-term after surgery
Response: according to your comments, we have supplemented that postoperative patients experienced pain and swelling leading to loss of range of movement, proprioceptive deficits, neuromuscular and biomechanical changes, decreased quadriceps femoris muscle strength, and further leading to dysfunction. The dysfunction especially in proprioception which perceives changes in the state of motion and initiates protection and muscular reflexes to keeps dynamic stabilizing were prone to increase the risk of degenerative disease or reinjury in long-term after surgery.

Line 47: The authors could make an effort to summarize more specifically which are the effects of proprioceptive training. Does evidence say that it is an intervention that could be applied as a single treatment for that condition? Is it better in combination with other interventions? What outcomes are expected to improve with this intervention?
Response: according to your comments, we have supplemented that adding proprioceptive training to conventional rehabilitation has been widely used in postoperative rehabilitation of joint at different times, some on the first day after surgery, and some three months after surgery, which proved to be effective in improving proprioception and activity function.

Line 47: In addition, paragraphs one and two should be better connected.
What deficits do those patients normally exhibit and why proprioceptive training may be the key to ameliorate those deficits? Please, explain better.
Response: according to your comments, we have made the modification.
.....The dysfunction especially in proprioception which perceives changes in the state of motion and initiates protection and muscular reflexes to keeps dynamic stabilizing are prone to increase the risk of degenerative disease or reinjury in long-term after surgery.
Proprioception plays an important role in postoperative rehabilitation. The inputs from the sensory organs are processed in the brain and integrated with visual and vestibular information to generate a sense of position and movement through space. Proprioceptive training can stimulate the sensory organ to send specific signals to control the relevant muscles to maintain stability through specific dynamic movement exercises such as practices of balancing, positioning, gait flexibility, agility and neuromuscular. In recent years, adding proprioceptive training to conventional rehabilitation has been widely used in postoperative rehabilitation of joint at different times, some on the first day after surgery, and some three months after surgery, which proved to be effective in improving proprioception and activity function.....

Lines 48-55: In my opinion, the need for this study could be better justified. Why may be proprioceptive training an appropriate intervention after meniscus arthroscopy? What are the expectations? How and when should it be performed?
Response: according to your comments, we have made the modification.
Proprioception training is beneficial to patients with APM. According to previous trials, some studies have shown that proprioceptive training has a significant improvement on knee rehabilitation for patients with APM. However, no significant difference is shown between patients undergoing proprioceptive training and counterparts without such treatment after surgery.
In recent years, adding proprioceptive training to conventional rehabilitation has been widely used in postoperative rehabilitation of joint at different times, some on the first day after surgery, and some three months after surgery, which proved to be effective in improving proprioception and activity function.

Page 4:

Line 7: Topic #4 of the PRISMA Checklist is not clearly shown in the manuscript. Please, define better the objective according to PIRSMA checklist description
Response: according to your comments, we have made the modification.
This study will not only meta-analyze the effect of proprioceptive training on the recovery of knee proprioception and muscle strength among patients with APM caused by traumatic tear, but also provide reliable medical evidence for the efficacy of proprioceptive training on the recovery of proprioception and function after APM.

METHODS

Line 29: Typographic mistake
Response: according to your comments, we have made the modification and checked the full text.

Line 45: Proprioceptive training may be also named "sensorimotor training". This keyword should be included in the search strategy.
Response: according to your comments, we have made the modification. And we also adjusted our search strategy.
Following keywords and their varies were used: proprioceptive training, sensorimotor training, proprioceptive neuromuscular facilitation, neuromuscular training, balance training, meniscectomy.

Page 5:

Line 5: Not really clear what the population inclusion criteria are for the present systematic review. As it is explained, it is not clear if it includes subjects who had undergone surgery for meniscus injury, or if it includes subjects who were operated for knee problems, and after that surgery, had developed meniscal injury.

In addition, in the introduction section, you correctly mentioned that there are two types of meniscus injury: one in young athletics subjects (traumatic tear) and the other in middle-aged subjects (degenerative tear). However, when you explain the population inclusion criteria, you do not specify which type of meniscus injury you are including. Considering the important differences that may exist in age, physical function and comorbidities between these two cohorts of subjects, it is not appropriate to make just a single analysis for these two groups. If you only include degenerative tears, please, clarify. If you combine both, make separate analyses.
Response: according to your comments, we have made the modification.
Population: patients who have undergone APM for meniscus injury caused by traumatic tear, whose race, nationality, and duration of disease are not limited, patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded.

Line 7: Please, explain better which intervention performs the control group. “Convectional care” may be misleading. Please, also clarify that control group intervention do not include proprioceptive exercises
Response: according to your comments, we have made the modification.
Comparison: conventional rehabilitation training (including ankle pump movement, continuous passive motion, weight bearing exercise, strength training, etc.) were applied to control groups, control groups intervention did not include proprioceptive training.

Line 3-25: Item # 6 of the PRSIMA Checklist is not fully done. Please, clarify this topic
Response: according to your comments, we have made the modification.
Eligibility criteria for this review based on PICOS frameworks (Population, Intervention, Comparison, Outcome and Study) were as follows: (1) patients who have undergone APM for meniscus injury caused by traumatic tear, whose race, nationality, and duration of disease are not limited, patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded; (2) Interventions: proprioceptive training alone or combined training with conventional rehabilitation were seemed as experimental groups; (3) conventional rehabilitation training (including ankle pump movement, continuous passive motion, weight bearing exercise, strength training, etc.) were applied to control groups, control groups intervention did not include proprioceptive training; (4) at least one of the outcomes of extensor, flexor muscle strength, knee function score or proprioception test was reported; (5) randomized controlled trial published in English or Chinese.

Line 33: At the beginning of the sentence, it is better to write the number in words, instead of figures
Response: according to your comments, we have made the modification and checked the full text.

Line 47: Is it better if the manuscript avoids first or second person
Response: according to your comments, we have made the modification and checked the full text.

It is not well explained what outcomes have been evaluated,
Response: Considering the comments of the editor and another reviewer. Since we used PEDro scale and The GRADE Approach, we deleted the
tool for risk of bias—the Cochrane Collaboration Network.

RESULTS

Page 6:

Line 49: It is better if you use the term “potentially relevant articles”, due to the fact that you should further assess if they are finally relevant
Response: according to your comments, we have made the modification.
As of December 2021, 155 potentially relevant literatures were retrieved according to the sampling criteria.

Page 7:

Line 19-25: In the methods section you mentioned that two researchers independently assessed the quality of the study using PEDRO scale. However, in Table 3 only appears a column of punctuations. Therefore, it is unknown which is the punctuation given by each of the researchers and if there was any disagreement between them. The same for risk of bias assessment.
Response: according to your comments, we have made the modification.
√: yes, no risk; ×: no, risky. This is also explained in the table note.

Lines 56-58: This sentence is not clear. Could authors rephrase it please?
Response: according to your comments, we have made the modification and checked the full text.
Five studies with 234 patients reported the flexor muscle strength of knee. The I2= 76% in Figure 3-a, with high heterogeneity.

Page 8:

Line 37: Clarify which tools have been used to test knee function. Are self-reported questionnaires or physical performance-based? Do all studies use the same approach?
Response: according to your comments, we have made the modification.
Five studies with 286 patients reported the function scores of knee assessed by Lysholm score and knee injury and osteoarthritis outcome score.

The selection process of the articles included in the review (Figure 1) is not in accordance with the inclusion and exclusion criteria exposed in the 'methods' section. You have excluded 1 record because "about anterior cruciate ligament injury" but there was no mention about it in 'methods'. Furthermore, it is not clear why you have excluded 5 articles "about meniscus injury". Please correct it.
Response: according to your comments, we have made the modification.
Patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded.
There were five studies in which patients had meniscus injury without surgery and proprioceptive training was used as a conservative treatment, so we excluded the 5 articles.

DISCUSSION

As a summary, the discussion section does not reflect the important points of the study (more explained in the commentaries below). Therefore, I suggest that authors should rephrase this section.
Response: according to your comments, we have rewritten the discussion. We suggest you to read this section in the revised manuscript.

I can read in the manuscript a lot of information (even maybe too much) about what is proprioception and how it works, but I miss more information regarding to how proprioception is affected in a meniscus injury and what is the impact of arthroscopic partial meniscectomy on proprioception.
Response: according to your comments, we have deleted a part of information about what is proprioception and how it works, and added information regarding to how proprioception is affected in a meniscus injury and what is the impact of arthroscopic partial meniscectomy on proprioception.

I consider that this review is supposed to be seeking if including proprioceptive training to conventional rehabilitation may improve knee function (both physical function and self-reported) over just conventional rehabilitation. However, this question is not resolved in the discussion.
Response: according to your comments, we have added this part in the discussion.
In addition, failure to improve knee function significantly in this study might be explained by poor transfer of the trained skills to gait. No exercises in the control group resemble the heel toe action of gait, and this degree of task specificity may be required to modify gait. Moreover, the setting of a blank control group and lack of assessment of compliance for two groups may further explain the less functional improvement observed in the proprioceptive training group.

It is not discussed if proprioceptive training may have clinical and functional implications on patients undergoing APM.
Response: according to your comments, we have added this part in the discussion.
In clinic, proprioceptive training is often added into conventional rehabilitation programs to improve the proprioception, muscle strength and function of patients after APM. Although the results of this

study show no significant improvement for proprioceptive training group in function score, proprioceptive training is still recommended in most studies to rehabilitation programs

I suggest to briefly compare the results obtained to other studies reviewing the effects of including proprioceptive training to conventional rehabilitation in other knee injuries
Response: according to your comments, we have added this part in the discussion.
In addition, proprioceptive training has been wide used in clinic, such as anterior cruciate ligament reconstruction, knee replacement, knee osteoarthritis, and have significant improvement in terms of the proprioception, rang of motion, balance, muscle strength, knee function, which shows that proprioception training is valuable for patients with knee disease to promote functional recovery.

Although it is little mentioned at the end of the limitations paragraph, you should discuss more about the dose, time, intensity, and type of exercises included in the proprioceptive interventions across all studies. Is it heterogenous? Do they lack some progressions? Do they do “enough” sessions?
Response: according to your comments, we have added this part in the discussion.
First, as seen in Figure 2, most of the included studies in this study had methodological flaws and small sample size, mostly the lack of participant, therapist and assessor blinding. These factors might overestimate the efficacy of proprioceptive training. More high-quality studies with larger sample should be implemented to confirm the efficacy of proprioceptive training. Second, the start time of intervention and assessment in the included studies was inconsistent. Study showed that the incidence of knee osteoarthritis was approximately 50% within 10 to 20 years from APM. This factor might affect the recovery of knee function and proprioception. In addition, this study generally focused on the effects of proprioceptive training after training and thus lacked studies assessing mid and long-term efficacy. The impact of follow-up time on efficacy of proprioceptive training needs to be explored further. Third, few studies were searched according the inclusion criteria, and different outcomes were also used. Therefore, the conclusions were not supported by sufficient data. More studies are needed to verify the conclusions of this study. Fourth, as seen in Table 1 and Figure 5, two studies had blank control groups that were different from those in other studies, which could possibly explain the heterogeneity observed in some outcomes. Due to the small number of studies, further subgroup analysis is not applicable. Future studies should standardize interventions in the control group. Finally, a lack of standardized exercise prescription needed to be noted, such as dose, time, intensity, and type of exercises, which would be the reason of high heterogeneity. Future studies should provide a standard and clear exercise prescription, which could help developing treatment and evaluation methods.

The response to reviewer 2

Reviewer: 2
Dr. Keng Lin Wong, National University of Singapore
Comments to the Author:
Overall, an important paper to discuss about post operative physiotherapy. The authors have did their best to consolidate the sparse literature, however, due to the limited amount of studies, the meta-analysis unfortunately is difficult to be considered as significant with the high I2. Meniscus repair versus partial menisectomy is likely the cause of the reason why the studies were so heterogenous.
Response: thank you for your comments concerning our paper. These comments are all valuable and very helpful for revising and improving our paper. We have read comments carefully and have made correction which we hope meet with approval.

I would suggest:
1) network meta-analysis of meniscus repair only or partial menisectomy only

2) if the results are too sparse then concentrate on the systematic review aspect only
and present both isolated meniscus repair or partial meniscectomy
Response: according to your comments, we have revised the inclusion criteria:
we only included patients who have undergone arthroscopic partial meniscectomy for meniscus injury caused by traumatic tear. Patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded. Our aim is to evaluate the effects of proprioceptive training on rehabilitation of knee after arthroscopic partial meniscectomy. A total of 9 studies were included in this study for meta-analysis, including 453 patients. There are indeed few studies that meet the inclusion criteria, and the low evidence supports the conclusion, it is necessary to carry out more samples and higher quality large-scale studies to provide high evidence in the future. In addition, we also discussed the importance of proprioception training for patients with arthroscopic partial meniscectomy in depth in the introduction and discussion section, and illustrated the positive effects of proprioception training applied to postoperative rehabilitation of other knee orthopedic.

The response to reviewer 3

Reviewer: 3
Dr. Abdelrahman Abushouk, Ain Shams University
Comments to the Author:
There are three major limitations that affect the reliability of the current meta-analysis findings. While the authors did attempt to mitigate them, their effects still apply.
Response: thank you for your comments concerning our paper. These comments are all valuable and very helpful for revising and improving our paper. We have read comments carefully and have made correction which we hope meet with approval.

1. The small number of included studies, which also have a small number of patients.
Response: constrained by inclusion criteria, small number of studies and patients was included. This is also illustrated in our limitations. Few studies were searched according the inclusion criteria, and different outcomes were also used. Therefore, the conclusions were not supported by sufficient data. More studies are needed to verify the conclusions of this study.

2. The high heterogeneity observed across most analyzed outcomes. Rightfully, the authors used random effects model and applied sensitivity analysis, when needed.
Response: we used random effects model and applied sensitivity analysis, when $I2 \geq 50\%$.

3. The additional major limitation is the different controls in the included studies, possibly explaining the heterogeneity observed in some outcomes. While some authors combine studies with different controls, it is not advised to do that and even when done, should be supplemented by subgroup analysis according to control type.
Response: This is illustrated in our limitations.
Fourth, as seen in Table 1 and Figure 5, two studies had blank control groups that were different from those in other studies, which could possibly explain the heterogeneity observed in some outcomes. Due to the small number of studies, further subgroup analysis is not applicable. Future studies should standardize interventions in the control group

I still have few comments:

1. In the methods section, the outcome measurements and definitions must be added, so that the readers understand what the results mean.
Response: according to your comments, we have made the modification.

Outcome: at least one of the outcomes of extensor, flexor muscle strength, knee function score or proprioception test was reported.

2. Also, it is imperative to define the intervention and mention how it was conducted in different studies, better done in a table.
Response: according to your comments, we have made the modification in the Table 1.

3. fixed effect NOT fixed effects.
Response: according to your comments, we have made the modification.

4. Page 6, line 37: What does it mean: The level of meta-analysis was set at .... Do you mean p value of significance?
Response: according to your comments, we have made the modification.
P value level of 0.05 was set for between-group differences.

5. The heterogeneity should better be interpreted according to Cochrane handbook cut-offs.https://handbook-5-1.cochrane.org/chapter_9/9_5_2_identifying_and_measuring_heterogeneity.htm
Response: according to your comments, we have made the modification.
Interpretation of I2 is as follows: 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity. When I2 was less than 50%, the combined effect was considered as mild heterogeneity and a fixed-effect model was utilized for meta-analysis; when I2 was greater than 50%, the heterogeneity was considered to be high.

6. Considering the limitations mentioned earlier, the authors are advised to perform GRADE assessment to evaluate the certainty of evidence.
Response: according to your comments, we have performed GRADE assessment to evaluate the certainty of evidence(Table 3).

7. Flexor Muscle Strength outcome: The significance here is also marginal (p= 0.05). Even if considered significant, its clinical value is questioned.
Response: according to your comments, we have made the modification.
By adopting random effects model, pooling of the data showed no statistical significance compared with control group (SMD= 0.56, 95% CI= [0.01, 1.11], P= 0.05). After sensitivity analysis, the study was identified as the source of high heterogeneity, which decreased significantly (I2= 0%, P= 0.04, Figure 3-b) after being excluded, which could justify the positive effects on knee flexor (hamstring) muscle strength brought by proprioceptive training.
Although the P value in improving flexor muscle strength is 0.05, the result is still valuable when the study that causes high heterogeneity is excluded, but further research is needed to confirm it.

8. Page 8, line 27: mention the exact p value.
Response: according to your comments, we have made the modification.
By using a fixed effects model, pooling of the data showed a statistical significance compared with control group (SMD= 0.31, 95% CI= [0.06, 0.56], P= 0.01), which could justify that proprioceptive training could improve knee extensor (quadriceps) muscle strength.

9. The figures for each outcome (before and after reducing heterogeneity) should be combined, making it easier for the reader.
Response: according to your comments, we have made the modification. We combine the figures for each outcome (before and after reducing heterogeneity)

10. In the "Evaluation of Knee Functions" outcome, the authors performed sensitivity analysis by removing two studies at a time. Not sure if this methodologically valid.
Response: In the "Evaluation of Knee Functions" outcome, we performed sensitivity analysis by removing one by one. And we explained it accordingly.

VERSION 2 – REVIEW

REVIEWER	Fernando Domínguez-Navarro University of Nottingham
REVIEW RETURNED	19-Dec-2021

GENERAL COMMENTS	Thanks to the authors for the time spent on addressing all the commentaries I made. I consider you made a great effort to improve the quality of the paper. Specially, introduction and discussion sections have gained more clarity, directness and a more clinically view. With this new version of the manuscript, most of my comments and concerns have been resolved satisfactorily. However, there are still some aspects that need to be clarify/improved 8appointed below) to go further in the review process. Furthermore, since it is obvious that English is not the native language for the authors, I suggest an English translating company (or someone with solvent use of written English) to review the language aspects of the manuscript. Abstract The results paragraph is not fully understandable. I suggest reviewing the writing Instead of “extensor muscle strength of knee” it is better saying: “knee extensor muscle strength”. The same for “flexor muscle strength of the knee” and “function score of knee”. Please correct it throughout the document. Strengths and limitations for the study I suggest the sentence to be rewritten like this: “This study is the first systematic review and meta-analysis to evaluate the functional effects of proprioceptive training after arthroscopic partial meniscectomy.” “Eight electronic databases were searched to provide a comprehensive range of studies.” Please, omit this point. Not relevant. Please, mention GRADE scores show very low quality Introduction With the corrections made, I consider right now Introduction to be sufficiently informative and accurate.

	Methods I consider this section be correct in the new version Results There exist contradictions in the manuscript between methods and results regarding exclusion criteria. In the methods section there says that patients with meniscus injury in combination with ACL injury requiring reconstruction would be excluded: “patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded”. However, in results sections in can be understood that two trials have not excluded those patients “Seven studies -from nine total studies- reported the exclusion of patients with combined ACL injuries,25 27 33-37 “. Please, review the participants characteristics for each study and made inclusion studies in accordance with what is exposed in methods. Otherwise, I suggest making separate analysis for those receiving ACL reconstruction in combination with APM, as they differ in functional condition and rehabilitation prognosis. I have experienced problems accessing to some of the articles included in the review. They cannot be found in Pubmed, as well as DOI number is absent too. I would like to ask authors to facilitate them in order to keep reviewing the manuscript. “Three studies with 141 patients reported the knee proprioceptive test.” I would suggest saying “Three studies, assessing 141 participants, performed knee proprioceptive test” After reviewing the age of the participants for some of the studies (Table 1), I am not completely sure all participants included in the review are traumatic meniscus tears, since patients older 40 years normally present degenerative tears. Please, check this item and consider using age as covariable in your analysis. In table 1, the mean participants’ age for the study Weiyan Huang 2018 is not correctly shown. Please, amend it. “Six studies with 259 patients reported the extensor muscle strength of knee.26-28 33 36 37 I2 =29% with low heterogeneity in Figure 4”. Those sentences are not clear, please, revise it. Discussion “It was found that the blinding and randomized sequential concealment was greatly biased while the incomplete data showed a minor bias” Please, omit this sentence “Partial meniscus impairment could also be found after APM. APM could not resulted in a significant improvement in proprioceptive function.” Avoid using the word APM so often in those sentences. “In addition, proprioceptive training has been wide used in clinic, such as anterior cruciate ligament reconstruction, knee replacement, knee osteoarthritis, and have significant improvement in terms of the proprioception, rang of motion, balance, muscle strength, knee function, which shows that proprioception training is valuable for patients with knee disease to promote functional recovery.” This

	sentence is too similar to what it is already exposed in introduction. Please, just say something like the findings obtained by this review reinforce what is found in other conditions (knee replacement, ACL reconstruction), suggesting proprioceptive to induce functional benefits. Conclusions It should be “based on very low quality” I suggest that, in this section, instead of talking about differences between groups for the evaluated, outcomes, you mention the implications. For instance: “including proprioceptive training may induce benefits for knee flexor strength and proprioception”.

REVIEWER	Abdelrahman Abushouk Ain Shams University
REVIEW RETURNED	08-Dec-2021

GENERAL COMMENTS	Although the current version is quite improved, the meta-analysis offers Very Low certainty evidence across all outcomes (as per GRADE assessment) that I do not believe can inform clinical practice.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1

-Thanks to the authors for the time spent on addressing all the commentaries I made. I consider you made a great effort to improve the quality of the paper. Specially, introduction and discussion sections have gained more clarity, directness and a more clinically view. With this new version of the manuscript, most of my comments and concerns have been resolved satisfactorily. However, there are still some aspects that need to be clarify/improved 8appointed below) to go further in the review process.

Response: Thank you for your comments concerning our paper. These comments are all valuable and very helpful for revising and improving our paper. We have read comments carefully and have made correction which we hope meet with approval.

-Furthermore, since it is obvious that English is not the native language for the authors, I suggest an English translating company (or someone with solvent use of written English) to review the language aspects of the manuscript.

Response: according to your comments, we carefully corrected words and phrases in our manuscript and our manuscript have been checked by someone familiar with writing article in English.

Abstract

-The results paragraph is not fully understandable. I suggest reviewing the writing

Response: according to your comments, we have rewritten the results paragraph.

-Instead of “extensor muscle strength of knee” it is better saying: “knee extensor muscle strength”. The same for “flexor muscle strength of the knee” and “function score of knee”. Please correct it throughout the document.

Response: according to your comments, we have made the modification and checked the full text.

Strengths and limitations for the study

-I suggest the sentence to be rewritten like this: “This study is the first systematic review and meta-analysis to evaluate the functional effects of proprioceptive training after arthroscopic partial meniscectomy.”

“Eight electronic databases were searched to provide a comprehensive range of studies.” Please, omit this point. Not relevant.

Please, mention GRADE scores show very low quality

Response: Because of the Comments from the Editor:“ This section should contain up to five short bullet points, no longer than one sentence each, that relate specifically to the methods. The novelty, aims, results or expected impact of the study should not be summarised here.”, we recalibrated this part.

Introduction

With the corrections made, I consider right now Introduction to be sufficiently informative and accurate.

Methods

I consider this section be correct in the new version

Results

-There exist contradictions in the manuscript between methods and results regarding exclusion criteria. In the methods section there says that patients with meniscus injury in combination with ACL injury requiring reconstruction would be excluded:

“patients who have undergone meniscal surgery combined with other procedures such anterior cruciate ligament reconstruction would be excluded”. However, in results sections in can be understood that two trials have not excluded those patients “Seven studies -from nine total studies- reported the exclusion of patients with combined ACL injuries,25 27 33-37 “. Please, review the participants characteristics for each study and made inclusion studies in accordance with what is exposed in methods. Otherwise, I suggest making separate analysis for those receiving ACL reconstruction in combination with APM, as they differ in functional condition and rehabilitation prognosis.

Response: A total of 9 original studies were included, among which 7 studies clearly described the exclusion of patients combined with ACL injuries, but the other two studies[26 28] did not report whether patients combined with ACL injuries were excluded in the methods or the full text. Given this, we have supplemented that the two studies[26 28] did not report whether patients combined with ACL injuries were excluded in the results section. Due to the limitations of the number of studies and different outcome indicators, it could not be independently analyzed. In addition, we have added this to the limitations, because the two studies[26 28] did not explicitly report that patients combined with ACL injuries were excluded, which may lead to potential heterogeneity, future studies should develop strict inclusion and exclusion criteria.

-I have experienced problems accessing to some of the articles included in the review. They cannot be found in Pubmed, as well as DOI number is absent too. I would like to ask authors to facilitate them in order to keep reviewing the manuscript.

Response: We described in the method that we included both Chinese and English studies. English studies could be retrieved in PubMed, while Chinese articles may only be retrieved in Chinese literature databases (such as https://www.cnki.net/; https://www.wanfangdata.com.cn/index.html;), and the sources of the articles are also explained in the references. For the convenience of review, we directly provide the original text in the form of an attachment, and the file name is the serial number of the reference.

-“Three studies with 141 patients reported the knee proprioceptive test.” I would suggest saying “Three studies, assessing 141 participants, performed knee proprioceptive test”

Response: according to your comments, we have made the modification and checked the full text.

After reviewing the age of the participants for some of the studies (Table 1), I am not completely sure all participants included in the review are traumatic meniscus tears, since patients older 40 years normally present degenerative tears. Please, check this item and consider using age as covariable in your analysis.

Response: Thank you very much for your suggestions. We have taken your suggestions seriously. We didn't change for the following reasons: As you said, there is indeed a lot of literature showing that patients older than 40 years normally present degenerative tears, and often combined with cartilage damage, which is more common in patients with knee osteoarthritis. Again, we carefully reviewed the patient characteristics included in the original studies. None of the original studies reported whether the included patients had been diagnosed with a traumatic or degenerative meniscal tear before arthroscopic partial meniscectomy. More importantly, 4 the original studies reported that the range of ages of the included patients was 35-45[37], 30-50[27], 20-60[35], and 29-58[34], and patients with moderate or severe cartilage damage would be excluded in the methods. This may indicate that patients older than 40 years were included and whose preoperative diagnosis was more likely to be traumatic meniscus tears in the 4 studies.

Firstly, this makes that we could not make a subgroup analysis of age, because we did a secondary analysis. Second, patients older than 40 years in the included studies may also have been diagnosed with a traumatic meniscus tear, which is the same as our inclusion criteria. So we didn't change. But we add it to the limitations, Hopefully, future studies should develop strict inclusion and exclusion criteria.

-In table 1, the mean participants’ age for the study Weiyan Huang 2018 is not correctly shown. Please, amend it.

Response: according to your comments, we carefully reviewed the patient characteristics included in the original study. When reporting demographic characteristics, this paper only reported the range of age of the included patients, without reporting the mean and standard deviation of patients' age.

-“Six studies with 259 patients reported the extensor muscle strength of knee.26-28 33 36 37 I2 =29% with low heterogeneity in Figure 4”. Those sentences are not clear, please, revise it.

Response: according to your comments, we have made the modification and checked the full text.

Discussion

-“It was found that the blinding and randomized sequential concealment was greatly biased while the incomplete data showed a minor bias” Please, omit this sentence

Response: according to your comments, we have omitted this sentence.

-“Partial meniscus impairment could also be found after APM. APM could not resulted in a significant improvement in proprioceptive function.” Avoid using the word APM so often in those sentences.

Response: according to your comments, we have made the modification and checked the full text.

-“In addition, proprioceptive training has been wide used in clinic, such as anterior cruciate ligament reconstruction, knee replacement, knee osteoarthritis, and have significant improvement in terms of the proprioception, rang of motion, balance, muscle strength, knee function, which shows that proprioception training is valuable for patients with knee disease to promote functional recovery.” This sentence is too similar to what it is already exposed in introduction. Please, just say something like the findings obtained by this review reinforce what is found in other conditions (knee replacement, ACL reconstruction), suggesting proprioceptive to induce functional benefits.

Response: according to your comments, we have made the modification.

Conclusions

-It should be “based on very low quality”

I suggest that, in this section, instead of talking about differences between groups for the evaluated, outcomes, you mention the implications. For instance: “including proprioceptive training may induce benefits for knee flexor strength and proprioception”.

Response: according to your comments, we have made the modification.